

Data Cleaning & Reporting Automation

Executive Pipeline Report | Generated on August 26, 2026 at 22:56:44

1. Executive Summary & Health Scorecard

Health Metric	Raw Ingestion	Cleaned Result	Pipeline Impact
Data Quality Score	71.2/100 [B (Acceptable)]	89.6/100 [A (Good)]	+18.4 Quality Gain
Total Rows	163	151	12 Duplicates Removed
Total Columns	10	19	Standardized snake_case
Missing Values	149	378	100% Imputed / Resolved

2. Key Performance Indicators (KPIs)

KPI Indicator	Computed Value	Description
Total Records	151	Total active rows in processed dataset
Total Revenue / Sales	\$81,405,053.55	Cumulative sum of 'net_revenue'
Average Order Value	\$539,106.31	Average ticket size from 'net_revenue'
Max Transaction	\$44,950,000.00	Highest recorded value in 'net_revenue'
Total Units Sold	8,113	Total volume from 'quantity'
Unique Customers / Accounts	24	Distinct entities in 'customer_name'

3. Visual Performance Analysis

Visual chart rendering skipped: name 'np' is not defined

4. Automated Insights & Findings

- **Insight:** Overall Data Health Score improved from 71.2/100 [B (Acceptable)] to 89.6/100 [A (Good)].
- **Insight:** Resolved -229 missing values across 10 columns using statistical imputation strategies.
- **Insight:** Eliminated 12 exact duplicate rows, preventing skewed metrics and double-counting.
- **Insight:** Successfully standardized data types for 3 columns (including currency symbol stripping, date parsing, and boolean normalization).
- **Insight:** Standardized 10 column headers into consistent snake_case format for SQL/pipeline compatibility.
- **Insight:** Total cumulative revenue / sales volume stands at \$81,405,053.55 across 151 transactions.
- **Insight:** Average Order Value (AOV) was calculated at \$539,106.31 per transaction.
- **Insight:** Top contributing category is 'Electronics' accounting for 69.6% of total volume (\$56,668,518.91).
- **Insight:** Noticeable concentration risk: 'Electronics' commands over half of the entire dataset revenue.
- **Insight:** Regional distribution is led by 'West' generating 61.1% of activity.

5. Strategic Recommendations

- **Action:** Implement frontend form validation at data entry to mandate required fields and eliminate downstream missing values.
- **Action:** Enforce database UNIQUE constraints on transactional ID fields to prevent duplicate records at ingestion.

- **Action:** Enforce strict schema typing at the ingestion pipeline to avoid storing raw currency symbols and unformatted strings in numeric fields.
- **Action:** Evaluate marketing and promotional strategies for lower-performing category 'Clothing' (4.2% share) to balance portfolio revenue.
- **Action:** Schedule automated daily/weekly runs of this cleaning pipeline to monitor data health drift over time.